

Experiment No. 4

Title: On Page SEO Optimization

Batch: B-1

Roll No.: 16010422234

Experiment No.: 4

Aim: Perform On Page SEO Optimization.

Resources needed: Internet, any On Page SEO Optimization tool and MS-office

Theory:

On-page SEO is the practice of optimizing individual web pages in order to rank higher and earn more relevant traffic in search engines. On-page refers to both the content and HTML source code of a page that can be optimized, as opposed to off-page SEO which refers to links and other external signals.

Before starting the on page optimization, keyword research needs to be carried out.

Steps for Keyword Research

- 1) Getting started with a brainstorm list
- 2) Acknowledging that you need a keyword research tool
- 3) Refining your list using suggested keyword phrases from an analysis tool
- 4) Verifying keyword phrase relevance
- 5) Looking at search volume to determine consumer demand
- 6) Analyzing the competitive space to make sure you and the searcher think the keywords mean the same things, and to decide if the space is too competitive

Once the keywords are finalized then the page can be optimized accordingly. Search engines love unique and high-quality original content. If the content of a website is only the web pages copied from other websites, most likely it will not be indexed by search engines. In addition, the website content being continually updated is the most basic condition for survival and development of a website.

Content optimization is based on following factors

CONTENT		
Cq	QUALITY	Are pages well written & have substantial quality content?
Cr	RESEARCH	Have you researched the keywords people may use to find your content?
Cw	WORDS	Do pages use words & phrases you hope they'll be found for?
Ce	ENGAGE	Do visitors spend time reading or "bounce" away quickly?
Cf	FRESH	Are pages fresh & about "hot" topics?

Content pages are the meat of websites and are almost always the reason visitors come to a site. Ideal content pages should be very specific to a given topic—usually a product or an object—and be hyper-relevant.

Title tag

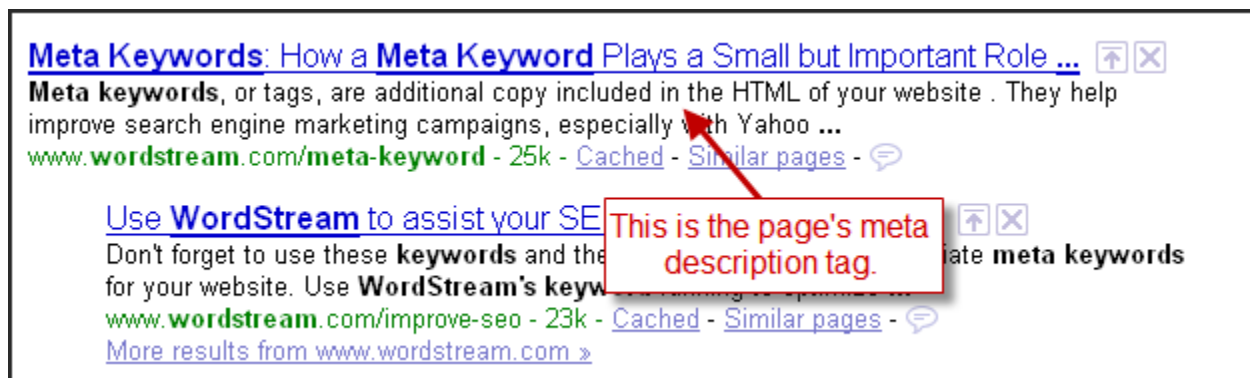
A title tag is an HTML element that specifies the title of a web page. Title tags are displayed on search engine results pages (SERPs) as the clickable headline for a given result, and are important for usability, SEO, and social sharing. The title tag of a web page is meant to be an accurate and concise description of a page's content.

Best Practices for Creating Titles

- Here are some best practices you should follow for creating titles on pages:
- Each page should have a unique title.
- If practical, try to include your Primary Keyword Phrase in every title of every page.
- Begin the title of your home page with your Primary Keyword Phrase, followed by your best Secondary Keyword Phrases.
- Use more specific variations to your Primary Keyword Phrase on your specific product, service, or content pages.
- If you must include your company name, put it at the end of the title.
- Use the best form, plural or singular, for your keywords based on what WordTracker says is searched on more often.
- Do not overdo it - do not repeat your keywords more than 2 to 3 times in the title.
- Make sure the <title> tag is the first element in the <head> section of your page - this makes it easier for Google to find the page.

Meta Description tag

The meta description is a useful meta tag as, very simply, it explains to search engines and (sometimes) searchers themselves what your page is about. Let's say you were googling the phrase "meta keywords" for example. You might encounter the following results:



The screenshot shows two search results from WordStream. The first result is titled "Meta Keywords: How a Meta Keyword Plays a Small but Important Role ..." and its meta description is "Meta keywords, or tags, are additional copy included in the HTML of your website. They help improve search engine marketing campaigns, especially with Yahoo ...". The second result is titled "Use WordStream to assist your SE ..." and its meta description is "Don't forget to use these keywords and the for your website. Use WordStream's keyw ...". A red box with a red arrow points to the meta description of the first result, with the text "This is the page's meta description tag." written inside the box.

Meta Keywords: How a **Meta Keyword** Plays a Small but Important Role ...  
Meta keywords, or tags, are additional copy included in the HTML of your website . They help improve search engine marketing campaigns, especially with Yahoo ...
www.wordstream.com/meta-keyword - 25k - [Cached](#) - [Similar pages](#) - 

Use **WordStream** to assist your SE  
 Don't forget to use these **keywords** and the for your website. Use **WordStream's keyw** 
www.wordstream.com/improve-seo - 23k - [Cached](#) - [Similar pages](#) - 
[More results from www.wordstream.com »](#)

This is the page's meta description tag.

Syntax-

```
<meta name="description" content="An accurate, keyword-rich description about 150 characters">
```

Meta Description Tag Tips

- Use keywords in your meta description tag.
- Try not to repeat the words overly often, but try to use multiple syntaxes of your key words.
- There should not be more than 150 characters in a description meta tag of a single web page.
- Use a different meta description tag for each page, as each page is different and stands a better chance of being found if you place a good title and description on it.

Meta keyword tag

The Keyword meta tag includes individual keywords separated by commas. Search engine crawlers use the Keyword meta tag information to index the website and display it in the search results. The keywords should be descriptive and optimized to ensure that the search engine displays the website when users search for these specific keywords.

```
<meta name="keywords" content="Keyword1, Keyword2, Keyword3">
```

Meta Robot tag

It explains to a spider of a search engine you only want it to index just the first page of your website or that it is allowed to index the whole website.

Syntax-

```
<meta name="robots" content="selection">
```

Ex-

```
<meta name="robots" content="index, follow">
```

- index/noindex - This tells the engines whether to show your page in search results or not.
- follow/nofollow - This tells the engines what to do with links on your pages: whether they should trust and "follow" your links to the next page or not.

Table 9-3. HTML meta directives

Directive	Description
Noindex	Instructs search engines <i>not</i> to index this page. Example: <code><meta name="robots" content="noindex" /></code>
Nofollow	Instructs search engines <i>not</i> to follow or drain any link juice for any outbound links. Example for all spiders: <code><meta name="robots" content="nofollow" /></code>
Nosnippet	Instructs search engines <i>not</i> to show any search results for this page. Example: <code><meta name="robots" content="nosnippet" /></code>
Noarchive	Instructs search engines <i>not</i> to show a cache link for this page. Example: <code><meta name="robots" content="noarchive" /></code>
Noodp	Instructs search engines <i>not</i> to use Open Directory Project descriptions in the SERPs. Example: <code><meta name="robots" content="noodp" /></code>
Follow	Default implied directive that says “follow” all outbound links. Example: <code><meta name="robots" content="follow" /></code>
Index	Default implied directive that says “index” this page. Example: <code><meta name="robots" content="index" /></code>

The content of a page is what makes it worthy of a search result position. It is what the user came to see and is thus extremely important to the search engines. As such, it is important to create good content. So what is good content? From an SEO perspective, all good content has two attributes. Good content must supply a demand and must be linkable. Duplicate content can present three main issues for search engines:

1. They don't know which version(s) to include/exclude from their indices.
2. They don't know whether to direct the link metrics (trust, authority, anchor text, link equity, etc.) to one page, or keep it separated between multiple versions.
3. They don't know which version(s) to rank for query results.
4. Thus content optimization is a major concern from SEO point of view.

Heading Tag optimization instructions

1. Never miss the H1 tag on a page as the search spiders tend to look for h1 tags to get the idea about the content of the page.
2. Don't overuse H1 tag, i.e. it should be only one for each page.
3. Use targeted keywords or phrases in the headings; avoid words that are not relevant to the page.
4. Use targeted keywords in the beginning of the headings.
5. The headings should be followed by a body of relevant content.
6. Maintain the order of heading tags; H1 should come first in the page then H2, H3 and so on.

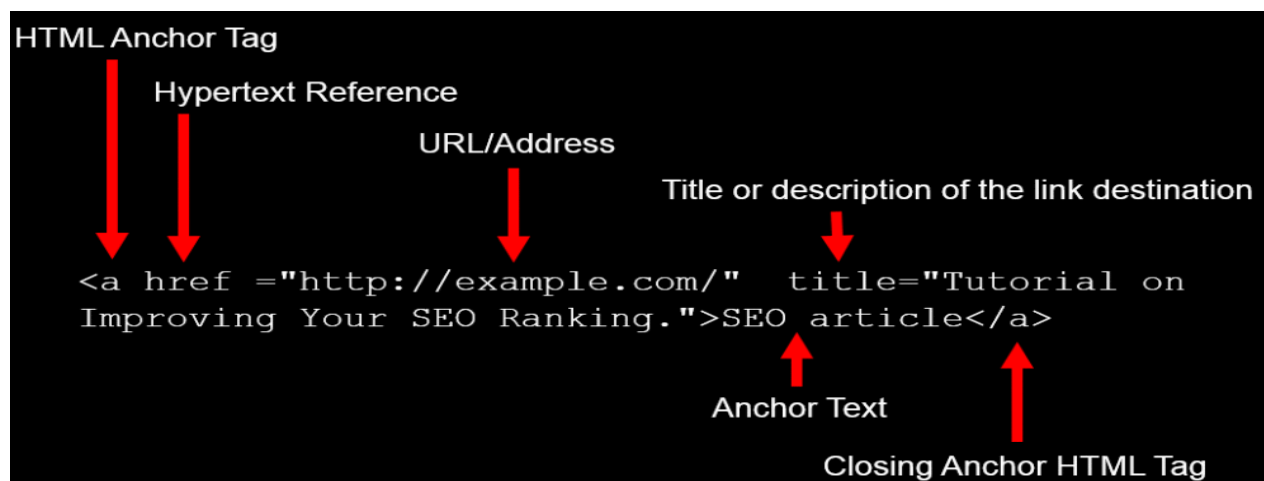
Images and Alt Text

```

```

- If an image fails to load or does not render on a page due to any reason then alt tag will appear to tell what the image is about. It helps users understand what kind of image was there.
- The search engines are designed to understand text on pages, not for the images. So, if your image is accompanied with alt tag the search engine will know from the alt tag what the image is showing and it will provide right images to the users.
- You can include your main keywords in the alt tag if they are related to the image. It will improve your image search ranking for that keyword.

Anchor text and link title



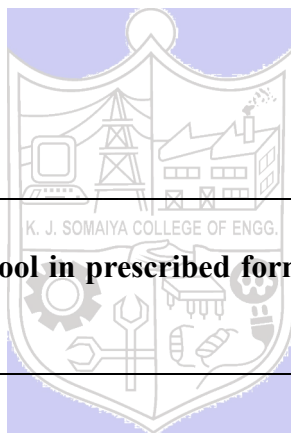
- Anchor text is the visible characters and words that hyperlinks display when linking to another document or location on the web.

- link TITLE is supposed to provide additional / advisory information about the link(expand on the meaning of the link).
- The anchor text is supposed to “name” the link, while the title text provides information about where the link will send the user. (especially with “click here” and “more” anchor text).

Procedure:

1. Perform on page SEO using online tools for any one page of your own website and prepare a report considering the following points in the following format.

- Keyword Optimization
- Title tag
- Meta description tag
- Meta keyword tag
- Meta robot tag
- Heading tag
- Anchor text
- Link title
- Alt Text in img tag



Results: (Description of selected tool in prescribed format. Prepare a separate table if the tool is different.)

Factor Under Consideration	On-page SEO
Name of the Tool (Include Company Name, website etc.)	Name of the Tool: SEMrush – Site Audit Tool Company: SEMrush Holdings, Inc. Website: https://www.semrush.com
Procedure	1. How does the tool accept the input? The tool accepts a website URL as input. The deployed site URL https://voluntr-volunteer-match-website.vercel.app was provided. 2. How does the tool process the data? (It is expected that you should include output screenshot of initial input and changes suggested by tool) SEMrush crawled 5 pages of the site and analyzed them for SEO issues related to metadata, content, links, and technical configurations. The tool produced a Site Health score and flagged issues in three categories: errors, warnings, and notices.

800 BOYLSTON STREET, SUITE 2475, BOSTON, MA 02199
WWW.SEMRUSH.COM

Site Audit: Overview

Subdomain: `voluntr-volunteer-match-website.vercel.app`
Last Update: August 11, 2025
Crawled Pages: 5

Site Health

74%

Crawled Pages

5

Healthy (1) Broken (0) Have Issues (3) Redirected (1) Blocked (0)

Errors

2

Warnings

7

Notices

2

Top Issues

Robots.txt file has format errors	errors	9% of total issues
1 issue with incorrect certificate name	errors	9% of total issues
1 page doesn't have enough text within the title tags	warnings	9% of total issues

Generated on August 11, 2025

The report data is taken from Semrush

2

3. Attach the result screenshot/changes made in code after incorporating changes.

i. Robots.txt file has format errors

Problem: The existing robots.txt was either missing, incorrectly formatted, or referenced a non-existent sitemap.xml.

Fix:

- Created a valid robots.txt file in the `/public/` folder with proper syntax and UTF-8 encoding.
- Linked it to a working sitemap.xml.

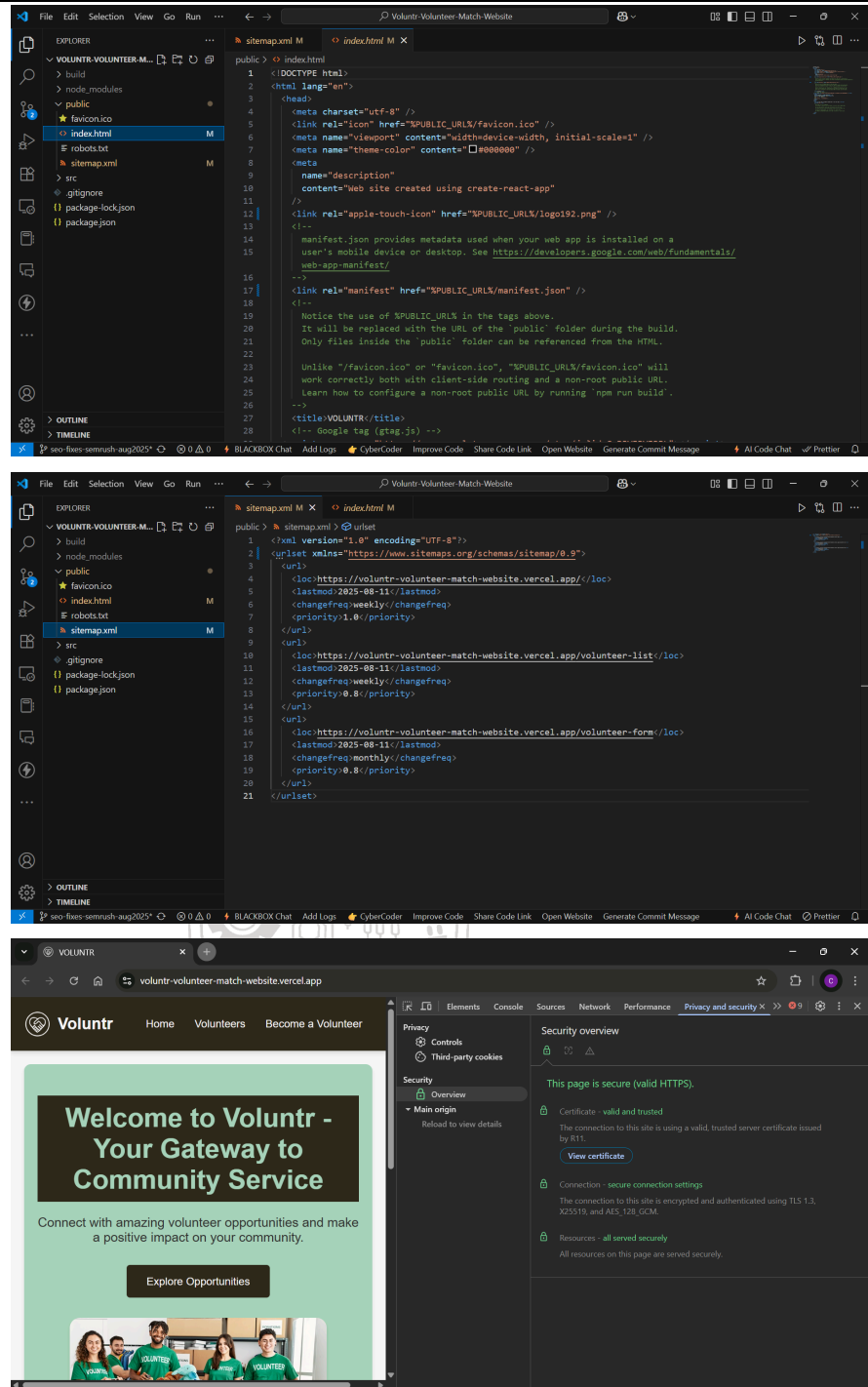
- Redeployed the site and confirmed the file loads at /robots.txt.

ii. Incorrect certificate name

Problem: Some resources (images, fonts, or scripts) were being loaded from domains whose SSL/TLS certificate did not match the hostname, or from http:// instead of https://.

Fix:

- Searched the codebase for all http:// references and replaced them with https://.
- Ensured all assets were loaded from correct domains matching the certificate's CN/SAN.
- Moved any mismatched third-party resources to /public/ and served them locally.
- Verified in Chrome DevTools → Security tab that there are no TLS mismatch warnings.



iii. Page doesn't have enough text within the title tag
 Problem: The home page had a very short and non-descriptive <title> tag, which reduced its relevance for search engines and made it less appealing in search results.

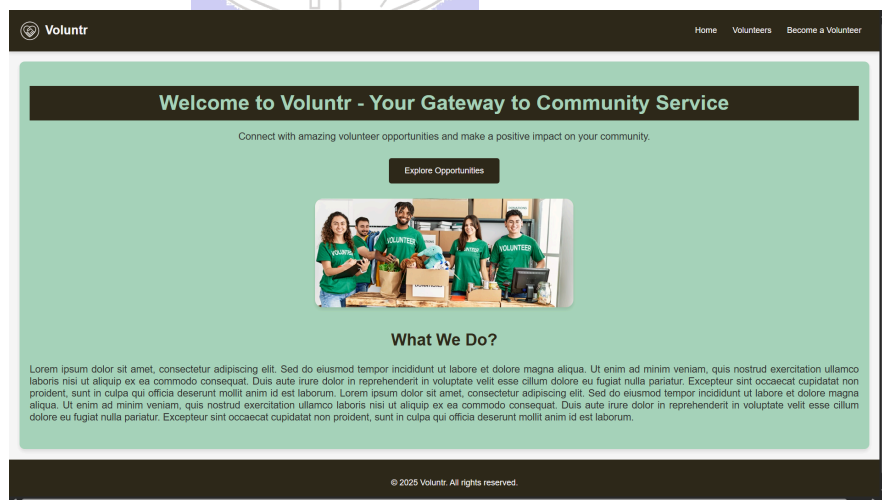
Fix:

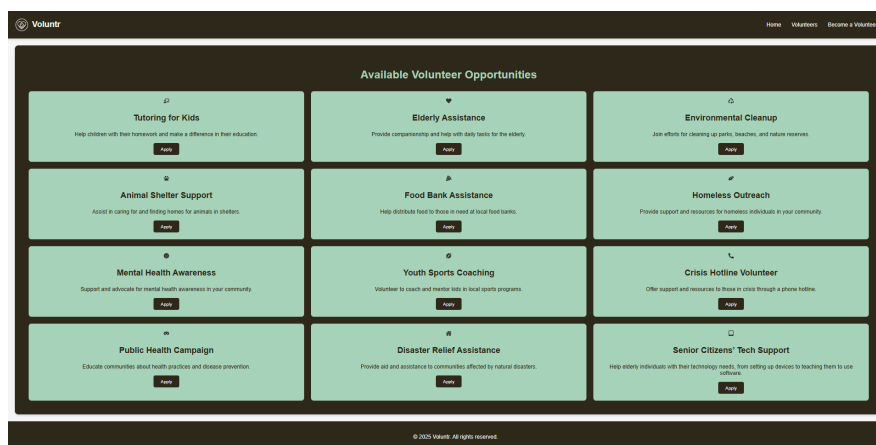
- Added a clear, keyword-rich <title> tag to public/index.html for the home page.
- Added a descriptive <meta name="description"> tag for better search result previews.
- Included meta keywords for additional SEO value as per project theory.
- Added meta robots to guide search engine crawling.
- Implemented Open Graph Card tags for improved social sharing previews, linking to a newly created image.png stored in /public/.

```

1  <html lang="en">
2  <head>
18  <!--
22  Unlike "/favicon.ico" or "favicon.ico", "/PUBLIC_URL/favicon.ico" will
23  work correctly both with client-side routing and a non-root public URL.
24  Learn how to configure a non-root public URL by running  npm run build .
25  -->
26  </head>
27  <title>Voluntr Match: Find Opportunities Near You | Voluntr</title>
28  <meta name="description" content="Discover verified volunteer opportunities by cause and city.
29  Filter and apply easily with Voluntr – start making an impact today.">
30  <meta name="keywords" content="volunteer match, volunteer opportunities, community service, find
31  volunteering roles, NGO volunteering, Voluntr">
32  <meta name="robots" content="index, follow">
33  <meta property="og:title" content="Volunteer Match: Find Opportunities Near You | Voluntr">
34  <meta property="og:description" content="Discover verified volunteer opportunities by cause and
35  city. Filter and apply easily with Voluntr – start making an impact today.">
36  <meta property="og:type" content="website">
37  <meta property="og:url" content="https://voluntr-volunteer-match-website.vercel.app/">
38  <meta property="og:image" content="/PUBLIC_URL/image.png">
39  <!-- Google tag (gtag.js) -->
40  <script async src="https://www.googletagmanager.com/gtag/js?id=G-BJVSRSVRPL"></script>
41  <script>
42    window.dataLayer = window.dataLayer || [];
43    function gtag(){dataLayer.push(arguments);}
44    gtag('js', new Date());
45    gtag('config', 'G-BJVSRSVRPL');
46  </script>

```





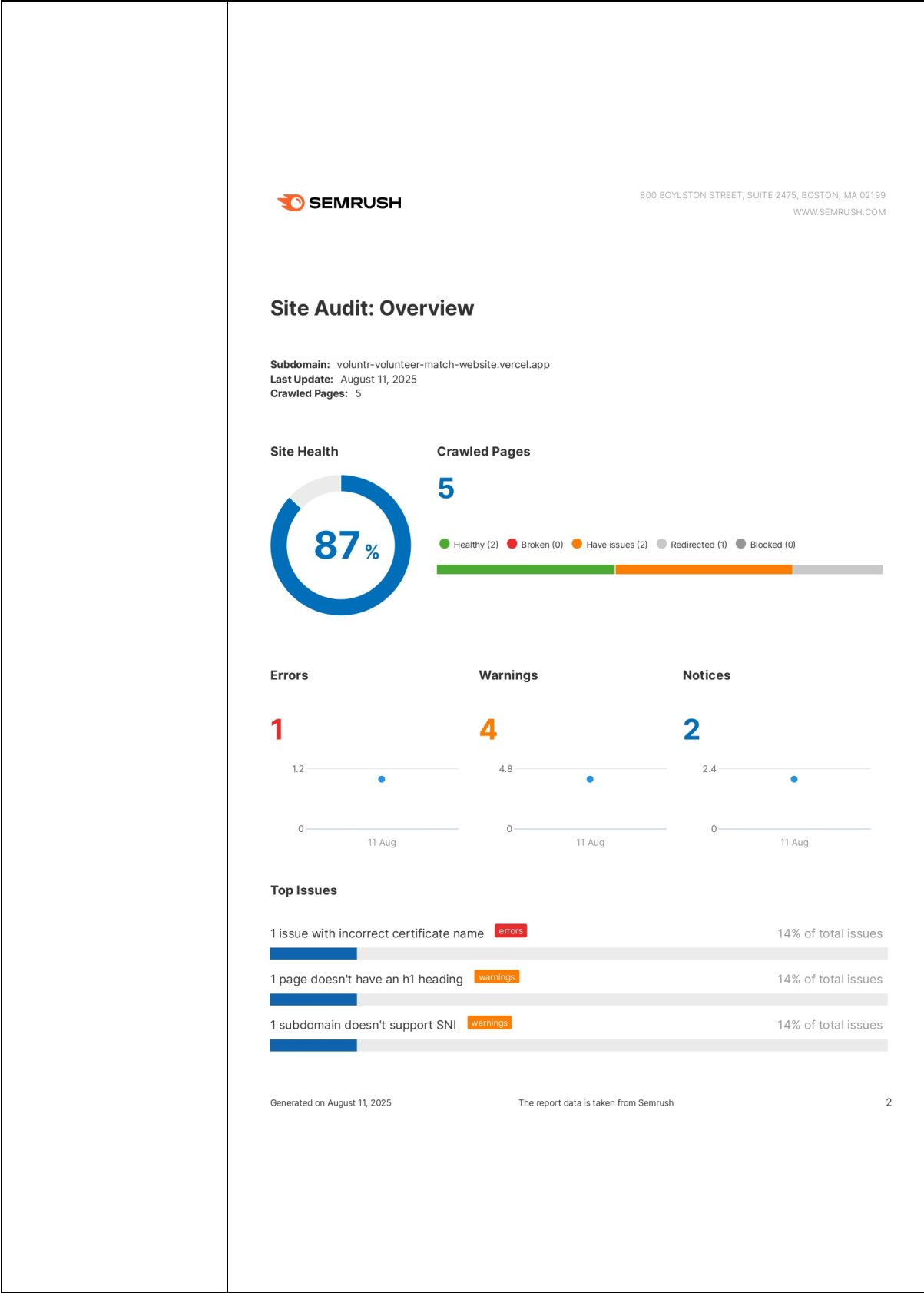
Voluntr

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Become a Volunteer

Submit

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Questions:

1. Explain off page SEO.

Ans: Off-page SEO refers to all optimization activities performed outside of your own website to improve its search engine ranking and visibility. Unlike on-page SEO, which focuses on elements within the website (content, HTML tags, structure), off-page SEO involves building the site's authority, trust, and popularity through external factors. Key off-page SEO techniques include:

1. Backlink Building – Earning high-quality links from other reputable websites to signal trust and authority to search engines.
2. Social Media Marketing – Sharing content and engaging with audiences on platforms like Facebook, Instagram, and LinkedIn to increase brand awareness and referral traffic.
3. Influencer Outreach – Collaborating with influencers or bloggers to get mentions and links.
4. Guest Posting – Publishing articles on relevant external sites with backlinks to your own.
5. Local SEO – Optimizing for Google Business Profile and citations in local directories.
6. Content Promotion – Distributing your content across various channels to reach a wider audience.

In essence, off-page SEO is about improving how the rest of the internet perceives and references your site.

Outcomes: CO2 – Understanding search engine optimization essentials

Conclusion: (Conclusion to be based on the objectives and outcomes achieved)

The on-page SEO optimization process successfully improved the overall search engine readiness of the selected webpage by implementing essential best practices such as proper keyword targeting, optimized meta tags, structured heading hierarchy, descriptive image alt attributes, and clean robots.txt configuration. These optimizations enhance both user experience and search engine visibility, resulting in better indexing, higher relevance for targeted keywords, and a stronger foundation for ongoing SEO improvements.

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of faculty in-charge with date

References:

1. Eric Greenberg, Alexander Kates, “Strategic Digital Marketing: Top Digital Experts Share the Formula for Tangible Returns on Your Marketing Investments” McGraw Hill Education 1st edition, 16 August 2013.
 2. Jan Zimmerman “Web Marketing For Dummies” Willy Publishing 3rd Edition, 2011.
 3. Jan Zimmerman, Deborah Ng, “ Social Media Marketing All-in-One For Dummies” Willy Publishing 4th Edition, 2017.
 4. Eric Enge, Stephan Spencer, Jessie Stricchiola, Rand Fishkin, “The Art of SEO”, 2nd Edition Mastering Search Engine Optimization O'Reilly Media 2nd Edition, 2012
 5. John I Jerkovic, “SEO Warriors” O'Reilly Media 1st edition, 2009
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